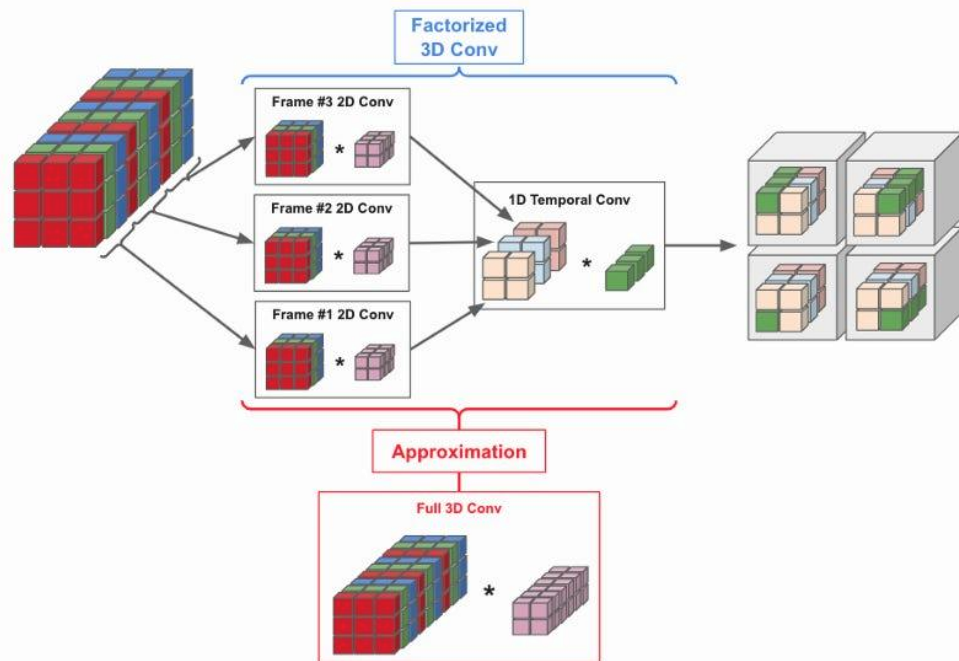


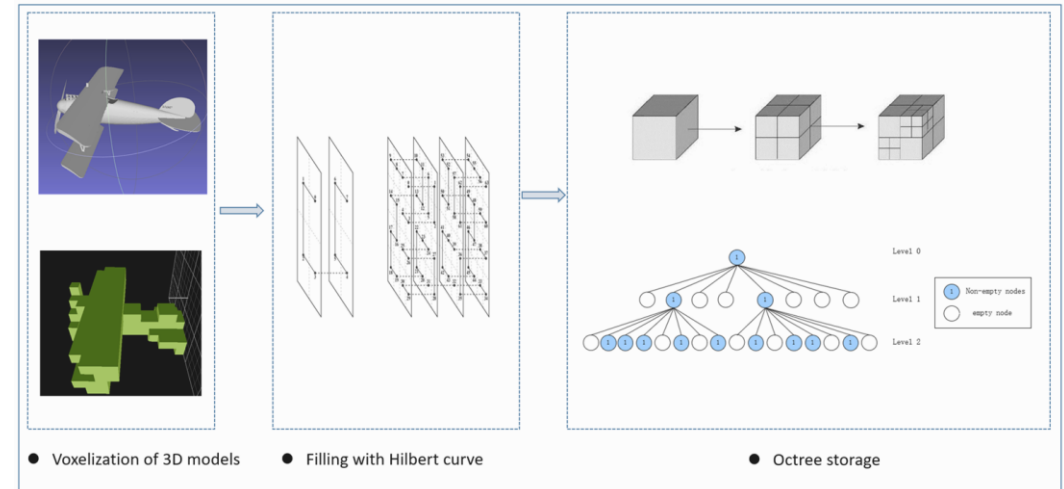


3D Convolutions: Beyond Flat Pixels

- Standard CNNs process images as 2D grids, but medical data like MRI or CT scans are volumetric.
- **3D Convolutions (Conv3D):** Instead of sliding a filter across height and width, the kernel slides across depth as well.
- This allows the model to capture spatial relationships across multiple slices, which is essential for detecting structures that span across a 3D space.

Voxel Analysis: Measuring the 3D World

- A **Voxel** is the 3D equivalent of a pixel (Volume + Element).
- **Automated Volume Calculation:** AI can be used to automatically calculate the volume of anomalies, such as tumors or lesions, by counting the predicted voxels.
- This provides a much higher level of precision for doctors compared to manual 2D measurements.

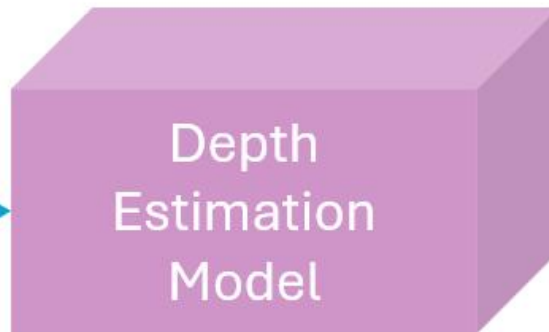


Depth Estimation: AI Understanding Distance

- Depth estimation is the process of determining the distance of objects from the camera in a 2D image.
- **Monocular Depth:** Predicting depth from a single image using patterns like perspective, shading, and object size.
- This technology is vital for autonomous driving (avoiding obstacles) and augmented reality (AR).



Input



Output